

Pockmark fields are locally ordered but not hyperuniform: a structure-factor analysis of Mediterranean seabed fluid-escape depressions

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Abstract

Submarine pockmarks — seabed depressions left by ascending fluids — occur in fields whose spatial arrangement has been described both as fault-controlled clustering and as regular spacing set by lateral inhibition. We ask a sharper, previously untested question: are pockmark fields *hyperuniform* — that is, do their long-wavelength density fluctuations vanish, as in avian photoreceptor mosaics and other disordered hyperuniform patterns — or is any regularity purely local? Using single fields drawn from a comprehensive open geodatabase of Mediterranean pockmarks, we combine the Clark–Evans nearest-neighbour index (local order) with the radially-averaged structure factor $S(k)$ and the number variance $\sigma^2(R)$ (long-range order), each compared against complete-spatial-randomness, hard-core (random sequential adsorption) and inhomogeneous-Poisson nulls. Local order spans the full range: one field is strongly clustered (nearest-neighbour index $R = 0.47$, below its heterogeneity null), one is essentially random ($R = 1.08$), and one is genuinely regular ($R = 1.26$, exceeding even a hard-core gas). Yet *every* field is long-range super-Poissonian — $S(k \rightarrow 0) > 1$ and $\sigma^2(R)$ growing with window area, not perimeter — and in each case the long-range behaviour matches the field’s own inhomogeneous-Poisson null. No field is hyperuniform; the single regular field is an anisotropic lineament, not an isotropic pattern. Pockmark order is therefore local and steric or structural, and is masked at long range by density heterogeneity. The result places pockmark fields with other self-organised geomorphic patterns whose local order does not extend to hyperuniformity, and shows that nearest-neighbour regularity must not be read as long-range order without a structure-factor test.

1 Introduction

Pockmarks are circular to elliptical seabed depressions formed where gas or pore water escapes through fine-grained sediment, first described on the Scotian Shelf (King & MacLean, 1970) and now known from continental shelves, slopes, fjords and lakes worldwide (Hovland & Judd, 1988). Within a field, individual pockmarks number from tens to many thousands, and the geometry of their spatial arrangement is a long-standing question. Two end-member pictures recur. In the first, pockmarks track subsurface fluid plumbing — faults, polygonal-fault networks, gas chimneys — and are correspondingly aligned or clustered. In the second, a field self-organises: each pockmark draws fluid from, and so depletes, a surrounding area, inhibiting the formation of neighbours within

a characteristic distance, so that the field acquires a regular, quasi-periodic spacing. This lateral-inhibition picture, and the use of nearest-neighbour and Ripley’s- K statistics to detect the resulting regularity, was set out by [Hammer \(2009\)](#) for a North Sea field.

A distinct and stronger notion of spatial order has meanwhile become central in condensed-matter physics and is increasingly applied to biological and geological patterns: *hyperuniformity*. A point pattern is hyperuniform if its large-scale density fluctuations are anomalously suppressed — formally, if the structure factor $S(k) \rightarrow 0$ as $k \rightarrow 0$, equivalently if the variance of the number of points in a window of radius R grows more slowly than the window’s area (as its perimeter, $\sigma^2(R) \sim R$, rather than $\sigma^2(R) \sim R^2$) ([Torquato & Stillinger, 2003](#); [Torquato, 2018](#)). Hyperuniformity is a *global* property: it cannot be read off local spacing. Disordered hyperuniform patterns occur in avian photoreceptor mosaics ([Jiao et al., 2014](#)) and a growing list of biological and physical systems, where they are thought to reflect an optimised balance of local irregularity and long-range uniformity.

The two notions are easily conflated. A pattern with a tight nearest-neighbour spacing — a hard-core liquid, or a field shaped by lateral inhibition — has a nearest-neighbour index well above the random value and looks “regular”, yet its long-wavelength fluctuations need not be suppressed at all: hard-core liquids are *not* hyperuniform. Distinguishing the two requires a long-range diagnostic (the structure factor or number variance), not a local one. Here we apply this distinction to pockmark fields for the first time. Using single fields from a comprehensive open geodatabase of Mediterranean pockmarks ([Spatola et al., 2025](#)), we ask whether the spacing regularity reported for such fields amounts to hyperuniformity, or is purely local order superimposed on a heterogeneous, fluid- and fault-controlled background.

2 Data and methods

Data. We use the geodatabase of [Spatola et al. \(2025\)](#), which compiles 7,516 georeferenced pockmarks across the Mediterranean and Black Seas from the published literature, each with up to 31 attributes, distributed as an ESRI point shapefile (projected coordinate system WGS84 UTM zone 33N, metres; licence CC-BY-4.0). Because the database is a multi-source compilation, we do not analyse it as a whole — pooling independent fields would trivially drive any statistic toward the uncorrelated (Poisson) limit. Instead we segment it by the `Location` attribute and select single fields that are (i) spatially contiguous, (ii) sufficiently populated, and (iii) free of the duplicate- and rounded-coordinate artefacts present in some entries. Three fields, spanning the range of local order, are analysed in detail: *Offshore Zannone Island* ($N = 112$), the *northwestern Calabrian margin* ($N = 418$), and the *Malta Plateau* ($N = 184$). All statistics are computed in the projected (metre) frame; for fields far from the central meridian the UTM scale factor varies negligibly across a single field.

Local order. We use the Clark–Evans nearest-neighbour index ([Clark & Evans, 1954](#)), $R = \bar{d}_{\text{NN}}/(1/(2\sqrt{\rho}))$, where $\bar{d}_{\text{NN}}$ is the mean nearest-neighbour distance and ρ the number density (estimated on the convex hull); $R < 1$ indicates clustering, $R = 1$ randomness, and $R > 1$ regularity, with a normal-approximation z -score.

Long-range order. We compute the radially-averaged structure factor

$$S(\mathbf{k}) = \frac{1}{N} \left| \sum_{j=1}^N e^{-i\mathbf{k} \cdot \mathbf{r}_j} \right|^2, \quad \mathbf{k} = 2\pi(n_x/L_x, n_y/L_y), \quad (1)$$

on the allowed wavevectors of the bounding box, and report its low- k limit $S(k \rightarrow 0)$ (the mean over $k < 0.4 k_\rho$, with $k_\rho = 2\pi\sqrt{\rho}$). We also compute the number variance $\sigma^2(R)$ from counts in random interior disks of radius R , and its log-log slope: a slope near 2 is area-like (Poisson or clustered), a slope near 1 is perimeter-like (hyperuniform).

Nulls. Each field is compared against (i) complete spatial randomness (CSR) in the same window; (ii) a random-sequential-adsorption (RSA) hard-core process with exclusion distance matched to the field’s nearest-neighbour scale, representing the steric / lateral-inhibition expectation; and (iii) an inhomogeneous-Poisson null that reproduces the field’s coarse density variations but is otherwise random, isolating the contribution of heterogeneity. A Gaussian-perturbed lattice provides a disordered-hyperuniform reference.

3 Results

3.1 Local order spans clustered to regular

The three fields differ markedly in local order (Table 1; Fig. 1). The northwestern Calabrian margin is strongly clustered ($R = 0.47$, $z = -20.9$), its points falling into several patches. Offshore Zannone Island is essentially random ($R = 1.08$, $z = +1.6$), a compact, near-isotropic cloud. The Malta Plateau is regular ($R = 1.26$, $z = +6.7$) but forms an elongated, diagonally-trending band (aspect ratio 2.5) — regular spacing along a lineament rather than an isotropic field.

3.2 The local order is genuine

Comparison against the nulls (Table 2) shows these local signals are real, not artefacts of window shape or density. The Calabrian nearest-neighbour index (0.47) lies below even its inhomogeneous-Poisson null (0.80): the field is more clustered than its density variation alone would produce, i.e. genuinely clustered into chains and patches. The Malta Plateau index (1.26) exceeds not only CSR (1.08) but also the hard-core RSA null (1.15): its spacing is more regular than a simple steric gas, consistent with an actively spaced (lateral-inhibition) lineament. Offshore Zannone tracks CSR (1.08 vs. 1.14), showing no resolvable local exclusion.

3.3 No field is hyperuniform

The long-range picture is uniform across all three fields and is the central result (Fig. 2, Fig. 3). Every field has $S(k \rightarrow 0) > 1$ — 2.5 (Zannone), 19.4 (Calabrian) and 5.4 (Malta) — and a number variance that grows with window *area* (log-log slope 2.8–3.3), not perimeter. The structure factor rises above unity at small k in every case and never shows the $S(k \rightarrow 0) \rightarrow 0$ suppression that defines hyperuniformity (Fig. 2b). In the order plane of local versus long-range order (Fig. 2a) the fields scatter widely along the nearest-neighbour axis but all sit high on the fluctuation axis; the hyperuniform region is empty.

Crucially, each field’s long-range fluctuation level matches its own inhomogeneous-Poisson null: $S(k \rightarrow 0) = 2.5$ versus 2.1 (Zannone), 19.4 versus 9.8 (Calabrian), 5.4 versus 6.7 (Malta). The excess density fluctuation is therefore largely the signature of density heterogeneity — patchiness, the elongated lineament, regional gradients — and not of any intrinsic long-range correlation. The only suppression of $S(k \rightarrow 0)$ below unity anywhere in the analysis appears in the hard-core RSA null (0.7–0.9), never in the data.

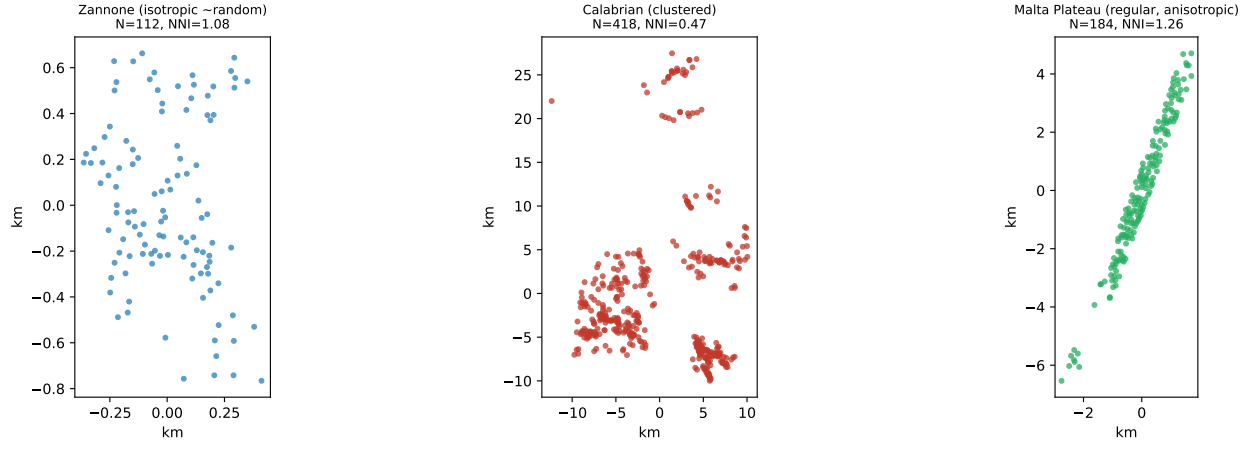


Figure 1: The three analysed pockmark fields in their projected (metre) frame, centred. Local order ranges from clustered (Calabrian, multi-patch) through near-random (Zannone) to regular but anisotropic (Malta Plateau, an elongated lineament).

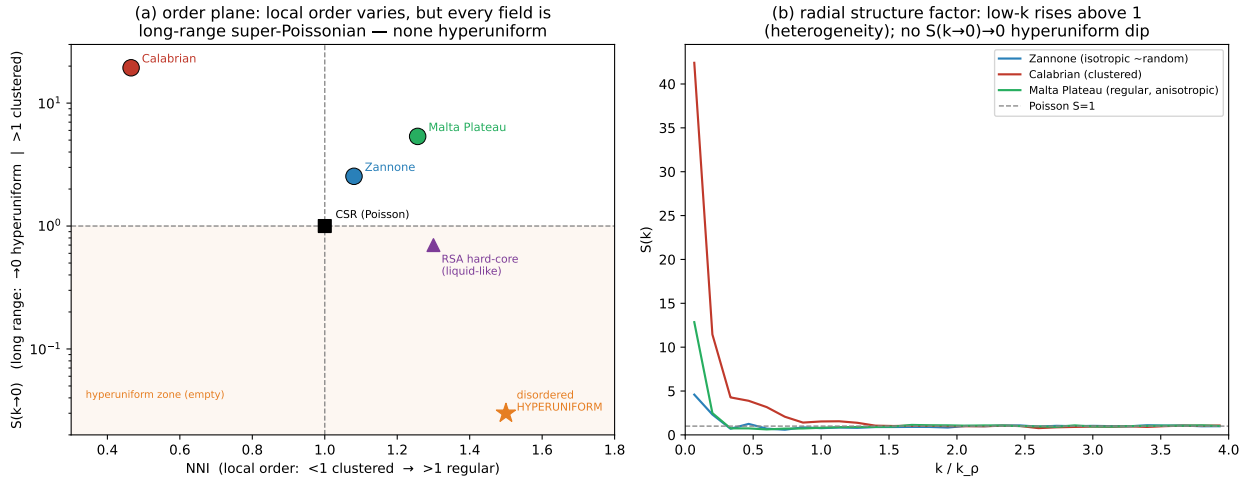


Figure 2: (a) Order plane: nearest-neighbour index (local order) versus $S(k \rightarrow 0)$ (long-range order, log scale). The fields scatter along the local axis but all lie well above the Poisson line on the fluctuation axis; the hyperuniform region (bottom, $S(k \rightarrow 0) \rightarrow 0$) is empty. Reference markers: CSR, a hard-core liquid (RSA), and a disordered-hyperuniform pattern. (b) Radial structure factor: every field rises above $S = 1$ at small k (density heterogeneity) and none shows the $S(k \rightarrow 0) \rightarrow 0$ dip of hyperuniformity. $k_\rho = 2\pi\sqrt{\rho}$.

Table 1: The three analysed pockmark fields. R is the Clark–Evans nearest-neighbour index (< 1 clustered, > 1 regular); d_0 the median nearest-neighbour distance; $S(k \rightarrow 0)$ the low- k structure factor; the last column the log–log slope of the number variance (≈ 2 area-like, ≈ 1 hyperuniform).

Field	N	extent (km)	aspect	R	z	d_0 (m)	$S(k \rightarrow 0)$ / NV slope
Offshore Zannone Island	112	1.4	1.8	1.08	+1.6	42	2.5 / 2.8
NW Calabrian margin	418	37.5	1.7	0.47	−20.9	211	19.4 / 2.9
Malta Plateau	184	11.2	2.5	1.26	+6.7	103	5.4 / 3.3

Table 2: Each field against its nulls: nearest-neighbour index R , low- k structure factor $S(k \rightarrow 0)$, and number-variance slope. Local order is real (data R departs from CSR in the expected direction and, for the clustered and regular fields, beyond the heterogeneity and hard-core nulls), but the long-range fluctuations of every field track its inhomogeneous-Poisson null — none is hyperuniform.

Field	Model	R	$S(k \rightarrow 0)$	NV slope
Offshore Zannone	data	1.08	2.54	2.81
	CSR	1.14	1.00	1.84
	RSA hard-core	1.31	0.72	1.84
	inhomogeneous-Poisson	1.04	2.05	2.60
NW Calabrian	data	0.47	19.4	2.90
	CSR	1.03	0.99	1.87
	RSA hard-core	1.07	1.02	1.93
	inhomogeneous-Poisson	0.80	9.77	3.32
Malta Plateau	data	1.26	5.36	3.28
	CSR	1.08	1.04	1.94
	RSA hard-core	1.15	0.91	1.93
	inhomogeneous-Poisson	0.94	6.73	3.20

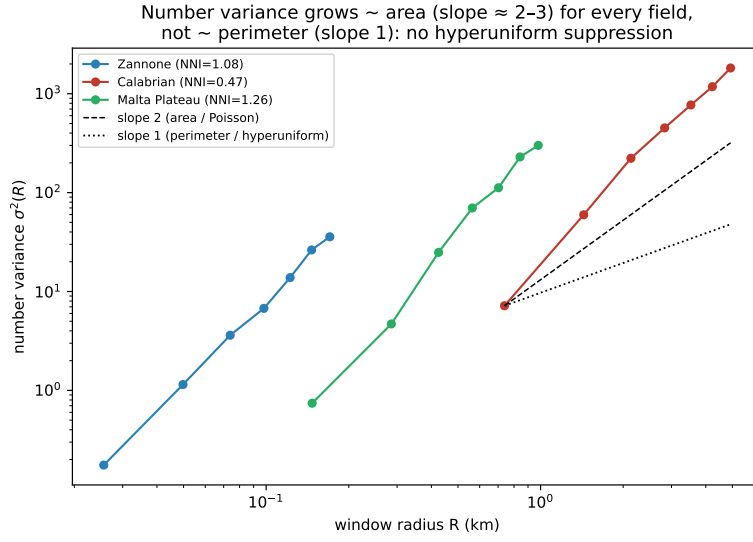


Figure 3: Number variance $\sigma^2(R)$ versus window radius. For every field $\sigma^2(R)$ grows with window area (slope $\approx 2-3$), parallel to the Poisson reference, rather than with perimeter (slope 1, the hyperuniform reference). No field shows hyperuniform suppression of density fluctuations.

4 Discussion

Across three Mediterranean pockmark fields spanning the full range of local order, the long-range verdict is unanimous: none is hyperuniform. Local regularity, where present, is genuine — the Malta Plateau is more regularly spaced than a hard-core gas, and the Calabrian margin is genuinely clustered — but it does not propagate to suppressed long-wavelength density fluctuations. Instead, the large-scale fluctuations of every field are super-Poissonian and are accounted for by density heterogeneity, as shown by their agreement with the inhomogeneous-Poisson nulls. This is the same local-versus-long-range structure seen in other self-organised geomorphic patterns: order that is real at the scale of neighbouring features but is overwritten at long range by the heterogeneity of the natural substrate.

The single regular field sharpens the point. The Malta Plateau’s high nearest-neighbour index might, on a local statistic alone, be read as evidence of an ordered, possibly hyperuniform, self-organised field. The structure factor and number variance show otherwise: the regularity is one-dimensional, expressed along an elongated lineament (aspect ratio 2.5), and the isotropically-averaged long-range fluctuations are if anything the largest of the three fields. Regular nearest-neighbour spacing of the kind attributed to lateral inhibition (Hammer, 2009) can thus coexist with strongly non-hyperuniform long-range behaviour; the two must be measured separately.

We are explicit about what is and is not new. That pockmark fields can be clustered along faults or regularly spaced by lateral inhibition is established (Hammer, 2009; Spatola et al., 2025), as is the regional, fault-related distribution of the Mediterranean database used here. The contribution of this note is the hyperuniformity-class characterisation — the structure factor, number variance and matched nulls — and the resulting verdict that the order in these fields is local and heterogeneity-limited, not hyperuniform. Several caveats bound the result. The database is a literature compilation of varying per-field completeness; we mitigated this by selecting contiguous, artefact-free fields, but completeness cannot be guaranteed, and incompleteness would, if anything, add spurious fluctuations rather than suppress them, so it cannot mask a true hyperuniform signal. The field sizes ($N \sim 10^2$) limit the lowest accessible wavenumber, so we claim only the absence of hyperuniformity, not a precise fluctuation exponent. Finally, the analysis is isotropically averaged; the Malta Plateau shows that directional analysis is the natural next step for lineated fields.

5 Conclusion

Submarine pockmark fields exhibit local spatial order ranging from clustered to regular, but a structure-factor and number-variance analysis of representative Mediterranean fields finds no hyperuniformity: long-range density fluctuations are super-Poissonian and governed by substrate heterogeneity, and the one genuinely regular field is an anisotropic lineament. Nearest-neighbour regularity in a fluid-escape field should therefore not be taken as evidence of long-range order without a structure-factor test.

Data and code availability

The pockmark geodatabase is from Spatola et al. (2025), Zenodo [10.5281/zenodo.15425907](https://zenodo.org/record/15425907) (CC-BY-4.0). Analysis code (a pure-Python shapefile reader, the point-pattern diagnostics, and the field-selection and figure scripts), the derived per-field coordinates and all result tables are available at <https://github.com/Ruqing1963/pockmark-field-order> and archived on Zenodo (DOI assigned on release).

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